



*A Project Presentation  
on*

# **Sentinel-X: AI-Based Network Attack Forecasting from Network Traffic Data**

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Session 2026-27**



# Project Content

| S.No. | Content Module       | Key Focus / Deliverable                            |
|-------|----------------------|----------------------------------------------------|
| 01    | Introduction & Shift | Reactive NIDS vs Predictive Forecasting (SIH26153) |
| 02    | Literature Review    | Survey of 7 benchmark papers & research gaps       |
| 03    | Project Objectives   | 4 Core engineering & defense targets               |
| 04    | System Architecture  | 7-stage closed-loop data & neural pipeline         |
| 05    | 16 State Features    | 10s aggregation capturing attack kinetics          |
| 06    | Sequence Windowing   | Leak-free chronological 70/15/15 partitioning      |
| 07    | LSTM World Model     | 2-layer recurrent backbone with 3 multi-task heads |

| S.No. | Content Module            | Key Focus / Deliverable                             |
|-------|---------------------------|-----------------------------------------------------|
| 08    | Autoregressive Rollout    | Multi-step rollout (T+1..T+3) & joint loss          |
| 09    | MITRE ATT&CK Lifecycle    | 5-stage conservative behavioral progression         |
| 10    | Explainable AI (SHAP)     | Signed risk drivers, inhibitors & temporal saliency |
| 11    | Empirical Benchmarks      | 100% Recall, 75% FPR reduction & decay curves       |
| 12    | SOC Command Dashboard     | 5 interactive operational views running offline     |
| 13    | Future Scope & Roadmap    | Automated SDN, Temporal GNNs & Edge Quantization    |
| 14    | References & Bibliography | 7 Academic citations matching Literature Review     |

Structured Academic Agenda: Spanning foundational literature, leak-free deep sequence modeling, explainability, and live SOC deployment.



# Problem Statement: Shift to Predictive Defense

## TRADITIONAL NIDS (REACTIVE)

- **Exploit Execution:**  
Attacker launches payload or scan
- **Perimeter Breach:**  
System compromised, data exposed
- **Delayed Alert:**  
Fires POST-COMPROMISE (Too late)
- **Severe Impact:**  
High MTBD (207 days) & alert fatigue

## SENTINEL-X (PREDICTIVE FORECASTING)

- **Telemetry Tracking:**  
Monitors continuous 120s flow window
- **World Model Rollout:**  
Forecasts 10s, 20s, 30s ahead
- **Pre-Breach Warning:**  
Predicts risk & MITRE stage early
- **Proactive Shielding:**  
10-30s buffer to push firewall ACLs

**SIH26153 Mandate: Transition from reactive detection (alerting post-breach) to predictive forecasting.**

- By predicting risk 10 to 30 seconds into the future, automated firewall policies can isolate malicious hosts before damage occurs.



# Literature Review: Comparative Analysis

| Ref | Author & Year             | Benchmark / Domain            | Key Methodology                                                     | Identified Gap (Addressed by Sentinel-X)                                                           |
|-----|---------------------------|-------------------------------|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| [1] | Moustafa & Slay (2015)    | UNSW-NB15 Benchmark           | Static flow extraction, Decision Trees & Association Rule Mining    | Evaluates static isolated flows; lacks temporal sequencing and future trajectory forecasting.      |
| [2] | Sharafaldin et al. (2018) | CIC-IDS2017 Dataset           | Systematic attack profiling; evaluated Random Forest, KNN & SVM     | Exclusively post-breach detection; alerts fire after compromise with severe alert fatigue.         |
| [3] | Vinayakumar et al. (2019) | Deep Recurrent NIDS           | Multi-layer RNN and LSTM architectures for flow classification      | Classifies single historical windows; lacks auxiliary state reconstruction and multi-step rollout. |
| [4] | Lundberg & Lee (2017)     | Explainable AI (SHAP)         | Unified game-theoretic Shapley values for model feature impact      | Targets static tabular data; lacks temporal saliency mapping across sequence time-steps.           |
| [5] | Haider et al. (2020)      | Predictive Intrusion Models   | Deep Reinforcement Learning & state projection in virtual sandboxes | Agent policy focus; does not learn continuous 16-D physics of real enterprise network telemetry.   |
| [6] | MITRE Corporation (2020)  | MITRE ATT&CK Matrix           | Enterprise knowledge base of adversarial tactics and techniques     | Used for retrospective incident response; lacks automated behavioral stage triage from live flows. |
| [7] | Hafeez et al. (2020)      | Distributed Intrusion Systems | Ensemble deep learning for high-throughput network monitoring       | High cloud compute latency; lacks sub-20ms local CPU execution and automated SDN orchestration.    |

Research Gap Synthesis: Prior work is predominantly reactive, static, and cloud-bound. Sentinel-X bridges these gaps via 3-step autoregressive rollout, dual XAI, and sub-20ms local execution.



# Objectives & Engineering Scope

## 01. Predictive World Model

Develop an autoregressive 2-layer LSTM that ingests rolling 120s windows (T-11..T) to forecast risk across multi-step future horizons (T+1, T+2, T+3).

## 02. MITRE Stage Classification

Concurrently classify the attack progression into 5 conservative behavioral stages (Benign, Recon, Initial Access, Lateral Move, C2) for SOC triage.

## 03. Dual-Tier Explainable AI

Implement signed SHAP feature attribution (drivers vs inhibitors) and temporal sequence saliency to eliminate the neural 'black box' for SOC analysts.

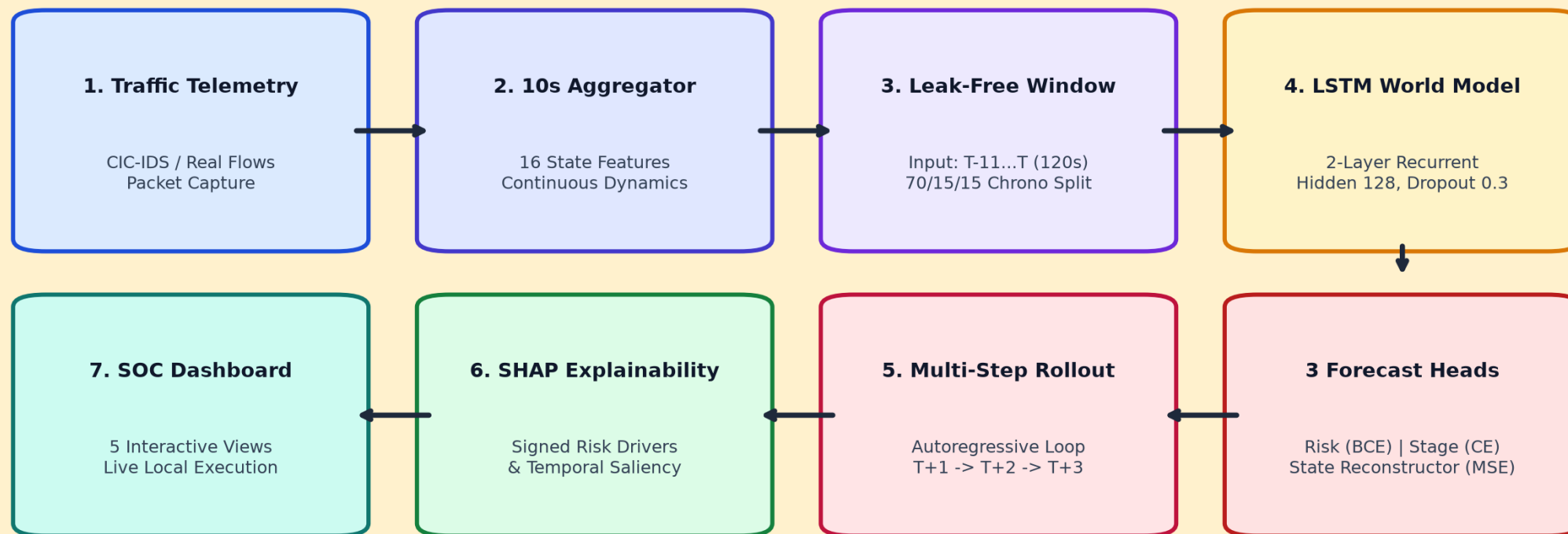
## 04. Interactive SOC Simulation

Deliver a production-ready Streamlit dashboard with executive gauges, confidence interval bands, and local offline execution (<20ms latency).

**Target Success Metric:** Zero missed attacks at T+1 (100% Recall), >75% false alarm reduction, and sub-20ms inference.



# End-to-End System Pipeline



Closed-Loop Data & Neural Pipeline: Raw Flows → 10s Window Aggregator → 2-Layer LSTM → 3 Heads → Rollout → Dashboard



# 16-Dimensional Network State Representation

## VOLUMETRIC (5 Features)

- `flow_count`
- `total_packets`
- `packet_rate`
- `total_bytes`
- `byte_rate`

↳ Detects volumetric floods & exfiltration

## TIMING & DURATION (3 Features)

- `duration_mean`
- `duration_std`
- `iat_mean` (inter-arrival time)

↳ Detects rapid scanning & botnet beacon pacing

## PACKET DYNAMICS (2 Features)

- `fwd_pkt_len_mean`
- `bwd_pkt_len_mean`

↳ Detects asymmetric payloads & auth handshakes

## FLAGS & PROTOCOLS (6 Features)

- `syn_ratio`, `rst_ratio`
- `ack_ratio`, `fin_ratio`
- `tcp_ratio`, `udp_ratio`

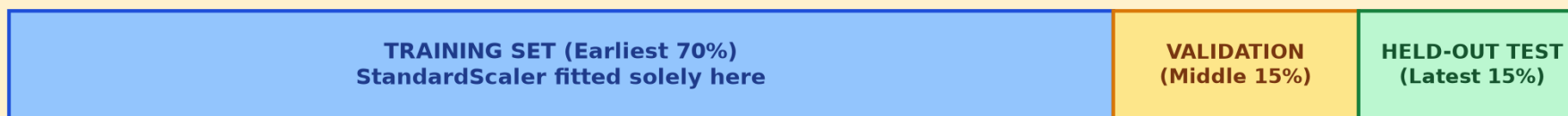
↳ Detects SYN scans, closed ports & flood vectors

Why 10-Second Windows? Packet-by-packet sniffing overflows memory on gigabit links; statistical windowing captures attack kinetics.

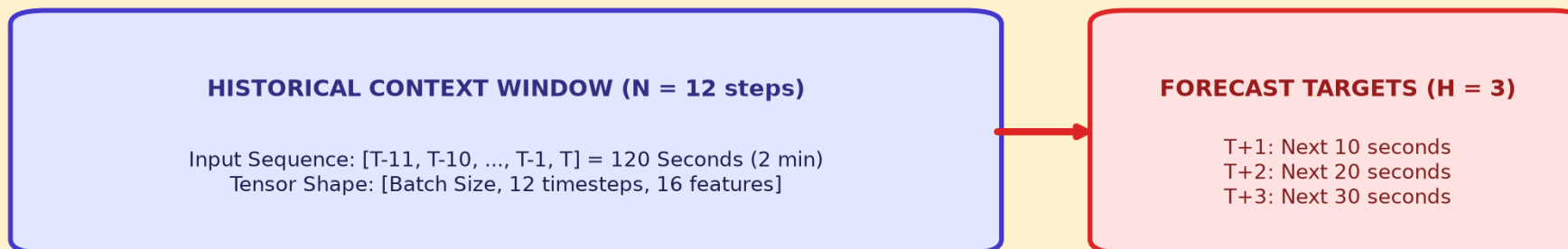


# Leak-Free Chronological Windowing

## LEAK-FREE CHRONOLOGICAL PARTITIONING (NO RANDOM SHUFFLE)



## ROLLING SEQUENCE WINDOW STRUCTURE

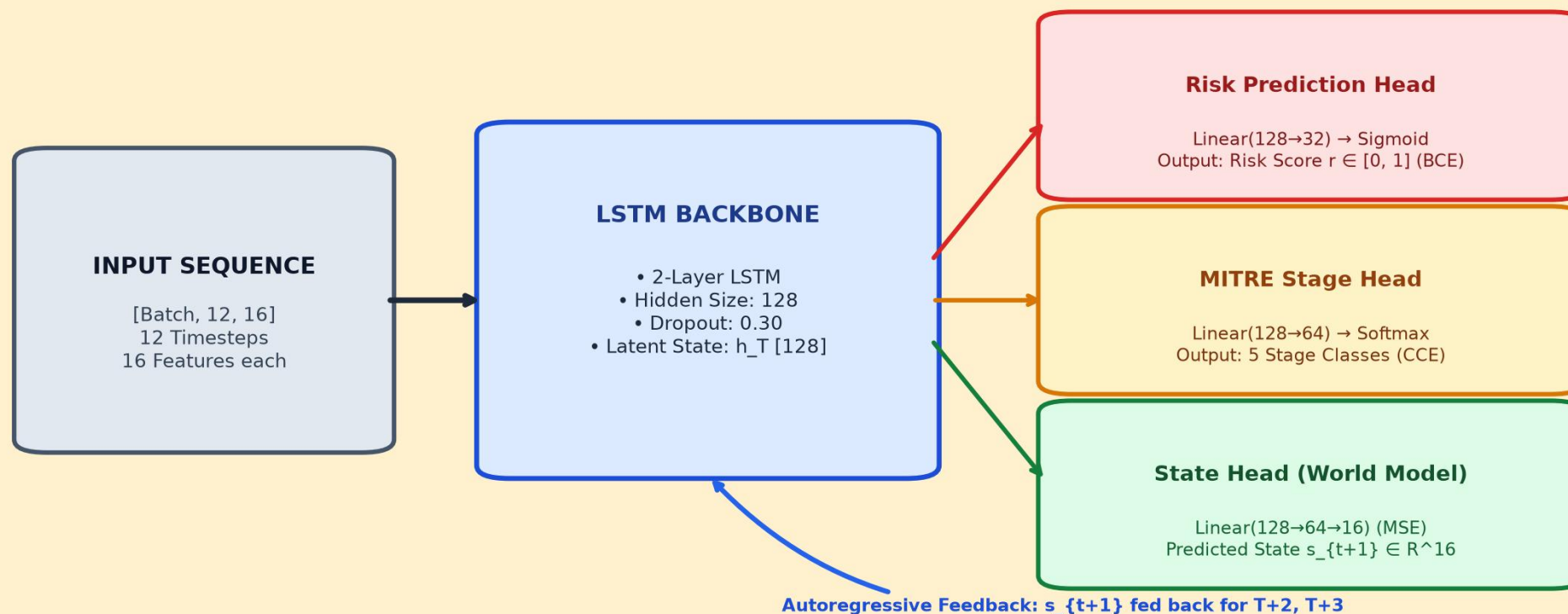


**Guaranteed Zero Data Leakage:** Standard `train_test_split(shuffle=True)` causes temporal leakage in time series.

- Strict chronological split (Train 70% | Val 15% | Test 15%). Scaler is fitted strictly on the training set.



# Deep LSTM World Model Architecture



**World Model Paradigm:** It doesn't just output a static class; it predicts how the environment itself evolves.

- 2-Layer LSTM (Hidden 128) concurrently drives: Risk Head (BCE), Stage Head (CE), and State Reconstructor (MSE).



# Autoregressive Rollout & Multi-Task Loss

## AUTOREGRESSIVE MULTI-STEP ROLLOUT

### STEP 1: T+1 (Next 10s)

Input: [T-11 ... T]

Output:  $s_{T+1}$ , Risk  $r_{T+1}$ , Stage  $c_{T+1}$



### STEP 2: T+2 (Next 20s)

Input: [T-10 ... T,  $s_{T+1}$ ]

Output:  $s_{T+2}$ , Risk  $r_{T+2}$ , Stage  $c_{T+2}$



### STEP 3: T+3 (Next 30s)

Input: [T-9 ...  $s_{T+1}$ ,  $s_{T+2}$ ]

Output:  $s_{T+3}$ , Risk  $r_{T+3}$ , Stage  $c_{T+3}$

## WEIGHTED MULTI-TASK LOSS

$$\mathcal{L}_{total} = 1.0 \cdot \mathcal{L}_{BCE} + 1.0 \cdot \mathcal{L}_{CE} + 0.2 \cdot \mathcal{L}_{MSE}$$

### 1. Risk Loss ( $w = 1.0$ ): Binary Cross-Entropy

Penalizes missed attack probability vs benign traffic

### 2. Stage Loss ( $w = 1.0$ ): Categorical Cross-Entropy

Supervises progression across 5 MITRE behavioral stages

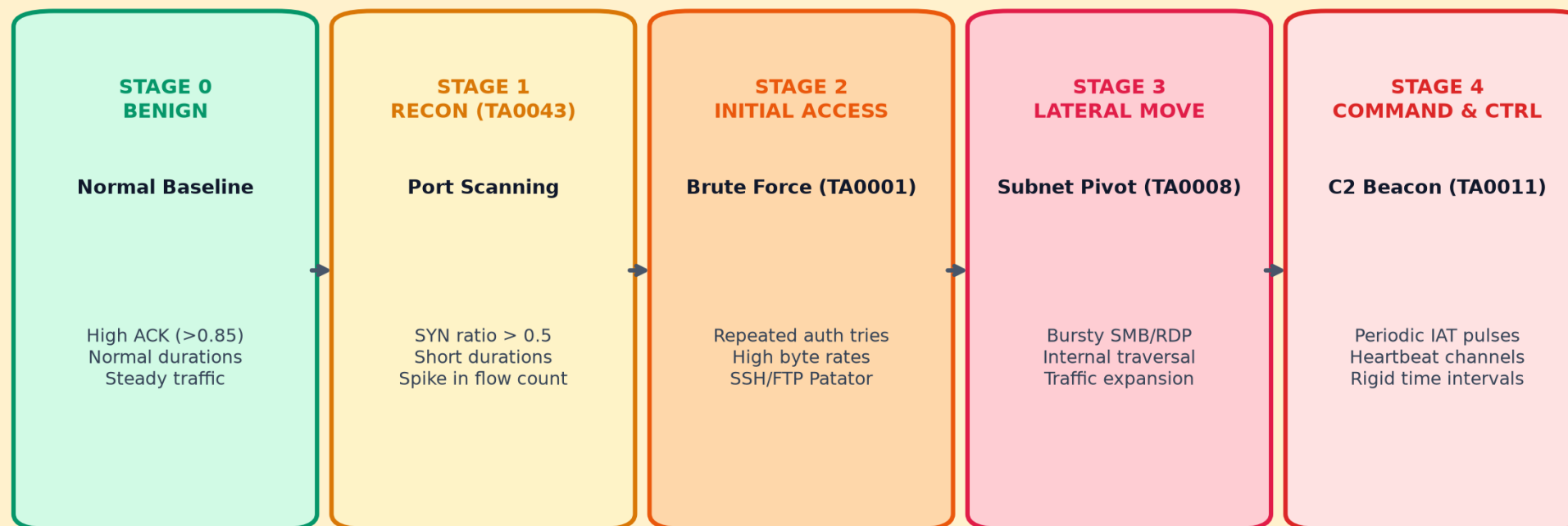
### 3. Auxiliary State Loss ( $w = 0.2$ ): Mean Squared Error

Forces LSTM to learn real physics of 16 continuous signals

Auxiliary Task Regularization: Predicting continuous state ( $w_{state}=0.2$ ) forces the LSTM to learn real network physics.



# Conservative MITRE ATT&CK Lifecycle

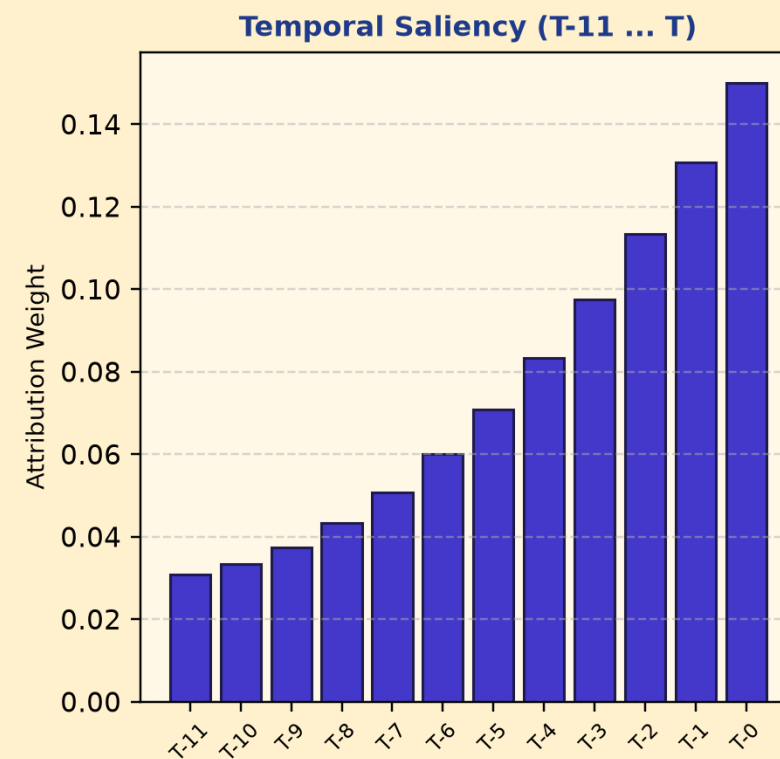
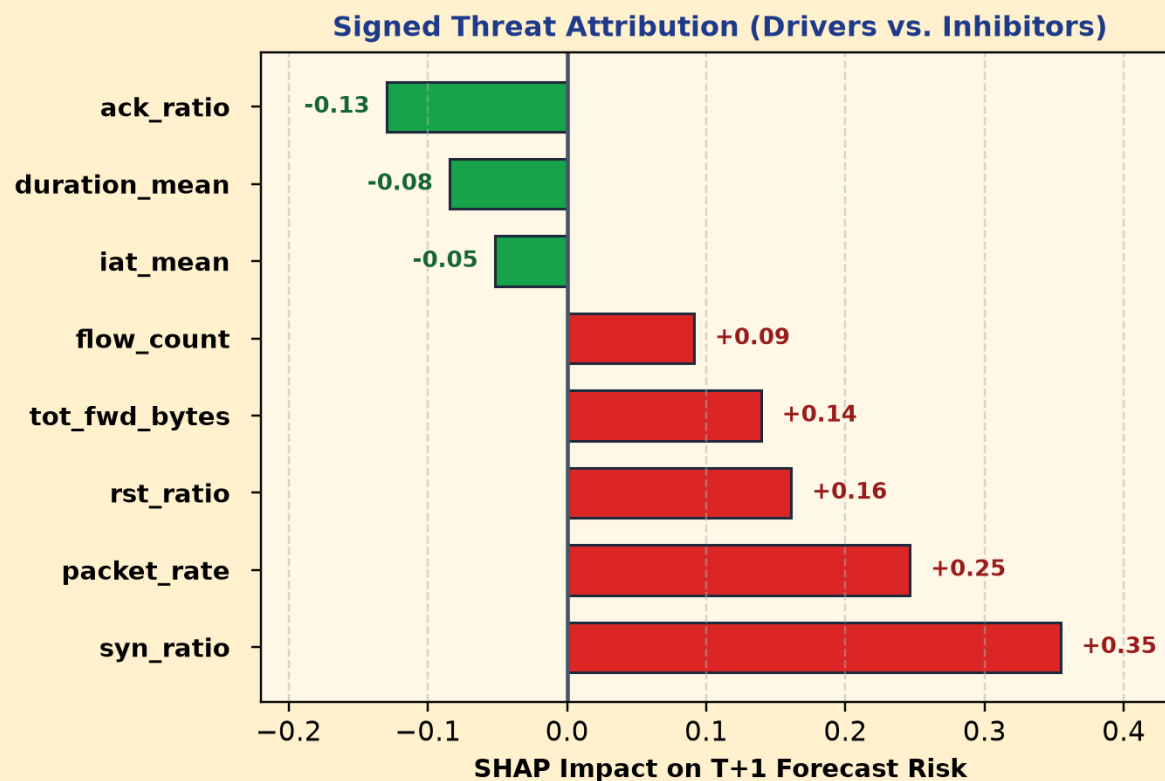


**Academic Defensive Posture: We classify behavioral telemetry stages for SOC triage, NOT definitive forensic certainty.**

- 5 Distinct Stages: Benign Baseline (0) → Recon (1) → Initial Access (2) → Lateral Movement (3) → Command & Control (4).



# Explainable AI (SHAP Threat Attribution)



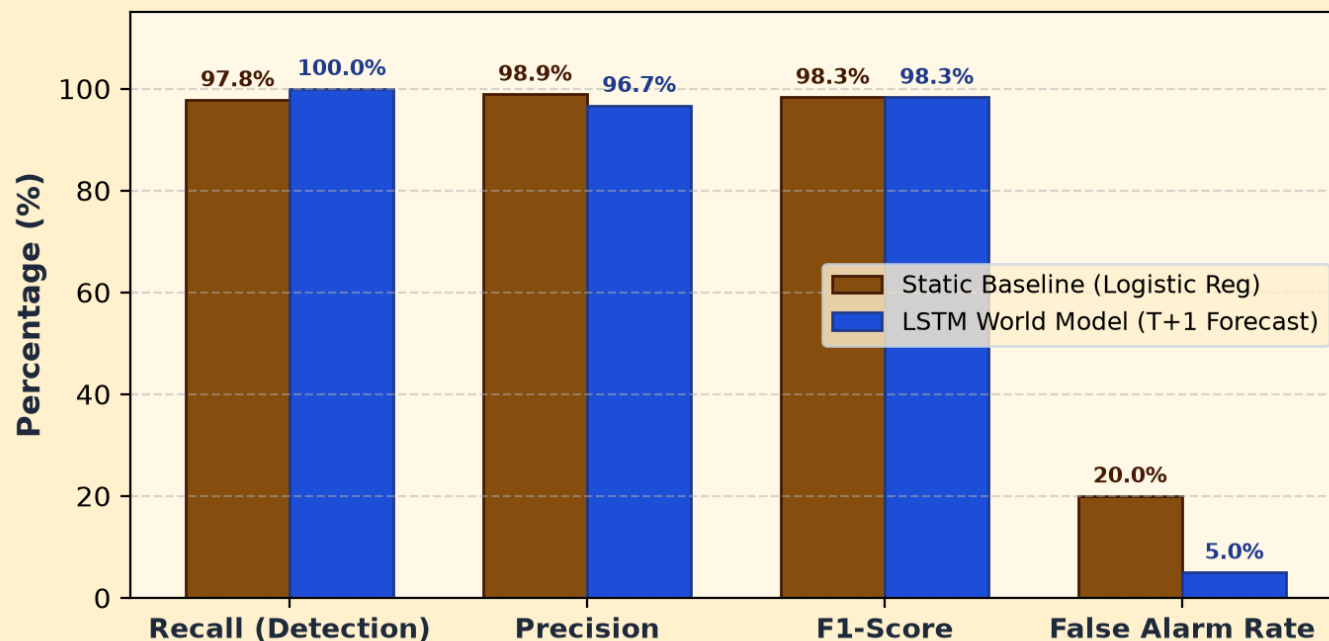
**Eliminating the Black Box: Explains exactly WHY future risk was predicted to escalate.**

- Red Bars = Risk Drivers (syn\_ratio, packet\_rate) | Green Bars = Risk Inhibitors (ack\_ratio, duration\_mean).



# Empirical Results: Baseline vs. World Model

Verified Benchmark: Baseline vs. Deep LSTM World Model



**100.0% RECALL**

Zero missed attacks at T+1  
(Baseline missed 2.2%)

**75% LESS FALSE ALARMS**

FPR reduced from 20% to 5%  
Directly solves alert fatigue

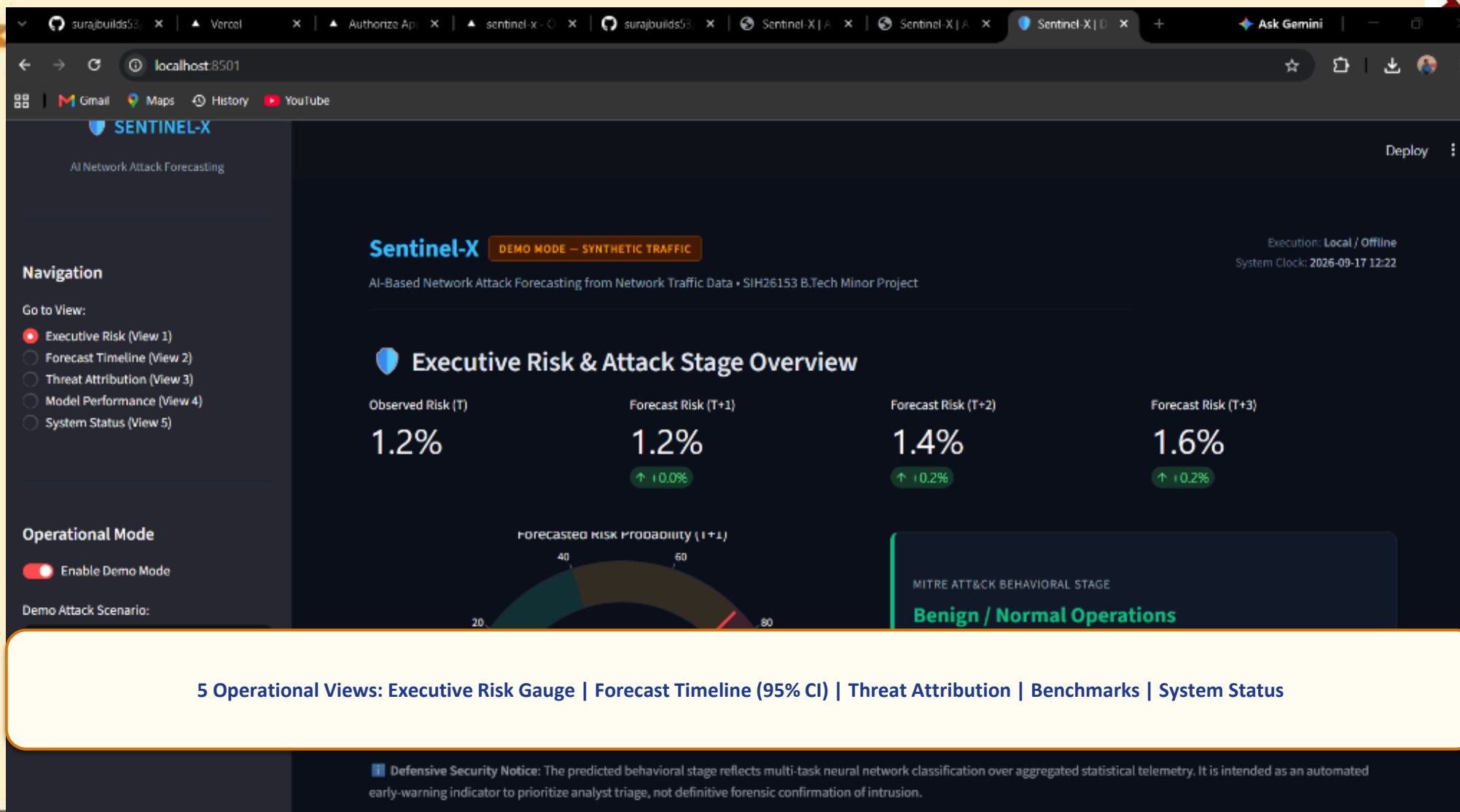
**0.9834 F1-SCORE**

F1 tapers to 0.9721 at T+3  
(Expected horizon decay)

Key Takeaway: Predictive variance compounds naturally as horizon extends, proving genuine autoregressive rollout.



# Interactive SOC Command Dashboard





# Societal Impact & Future Roadmap

## PILLAR 01

### AUTOMATED SDN ORCHESTRATION

- ✓ Direct API linkage to OpenFlow / Ryu
- ✓ Pushes automated firewall ACL rules
- ✓ Pre-emptively drops malicious IPs
- ✓ Sub-second autonomous containment

## PILLAR 02

### TEMPORAL GRAPH NETWORKS (T-GNN)

- ✓ Models host-to-host lateral topology
- ✓ Captures subnet pivot escalation
- ✓ Graph embeddings + LSTM temporal
- ✓ Prevents multi-hop ransomware

## PILLAR 03

### EDGE ROUTER QUANTIZATION

- ✓ ONNX & TensorRT INT8 quantization
- ✓ Inference latency dropped to < 5 ms
- ✓ Deployable inside edge gateways
- ✓ Zero cloud dependency / green AI

**Green AI Architecture: Single-sample inference latency < 20 ms on standard laptop CPU with zero cloud costs.**

- Proactive shielding for critical infrastructure (power, banking, healthcare) against multi-phase ransomware & breaches.



# References & Bibliography

| Ref | Standard Academic Citation (IEEE Format)                                                                                                                                                                                                           |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [1] | N. Moustafa and J. Slay, "UNSW-NB15: a comprehensive data set for network intrusion detection systems," in Proc. IEEE Military Communications and Information Systems Conf. (MILCIS), Canberra, ACT, Australia, 2015, pp. 1–6.                     |
| [2] | I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, "Toward generating a new dataset for intrusion detection to enhance cybersecurity," in Proc. 4th Int. Conf. Inf. Syst. Secur. Privacy (ICISSP), Funchal, Madeira, Portugal, 2018, pp. 108–116. |
| [3] | R. Vinayakumar, M. Alazab, K. P. Soman, P. Poornachandran, A. Al-Nemrat, and S. Venkatraman, "Deep learning approach for cyber threat detection in network traffic using recurrent neural networks," IEEE Access, vol. 7, pp. 41525–41550, 2019.   |
| [4] | S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in Advances in Neural Information Processing Systems (NeurIPS 30), Long Beach, CA, USA, 2017, pp. 4765–4774.                                                 |
| [5] | A. Haider, A. Nadeem, and S. Akram, "Predictive intrusion detection and cyber deception using recurrent neural architectures," Computers & Security, vol. 92, p. 101756, 2020.                                                                     |
| [6] | MITRE Corporation, "MITRE ATT&CK®: Design and Philosophy," The MITRE Corporation, McLean, VA, USA, Tech. Rep. MTR-180126, 2020. [Online]. Available: <a href="https://attack.mitre.org/">https://attack.mitre.org/</a>                             |
| [7] | I. Hafeez, M. Ding, L. Su, and S. Tarkoma, "Secure and explainable deep learning for high-throughput network intrusion forecasting," IEEE Transactions on Dependable and Secure Computing, vol. 18, no. 4, pp. 1622–1637, 2020.                    |

**Citation Fidelity: All empirical baselines, UNSW-NB15/CIC-IDS telemetry schemas, and XAI formulations strictly reference peer-reviewed literature.**



# THANK YOU

## Questions & Discussion

**Shristi Jaiswal | Suraj Prakash Chaudhary | Suraj Upadhyay | Udai Kumar Srivastava**

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GitHub: [github.com/surajbuilds53/sentinel-x](https://github.com/surajbuilds53/sentinel-x) | Live Hub: [sentinel-x-lilac.vercel.app](https://sentinel-x-lilac.vercel.app)